

Institute of Actuaries of India

Subject CT2 – Finance and Financial Reporting

October 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1: B. Accrual [2]

Solution 2: A. the proportion of equity of a company available for trading by the public on a stock market. [2]

Solution 3: C. Shareholders get tax relief on the dividend income as company has already paid tax on the profits earned. [2]

Solution 4: B. 30%% [2]

Solution 5: C. II only [2]

Solution 6: B. I & II [2]

Solution 7: C. I & III [2]

Solution 8: C. Both are equal. [2]

Solution 9: A. I, II & III [2]

Solution 10: D. I only [2]

Solution 11: i)

Income statement for the financial year ended on 31st March 2014

	INR million	INR million	Marks
Revenue		19000.00	1
Cost of sales		14600.00	1
Opening inventory	350.00		0.5
(+) Purchases	12000.00		1
(+) Depreciation on machinery	1335.00		1
(+) Manufacturing expenses other than depreciation on Machinery	1965.00		0.5
(-) Closing inventory	1050.00		2
Gross profit		4400.00	1
Administrative expenses & other distribution cost		3400.00	
(-) Depreciation on office building	37.50		1
(-) Administrative expenses & other distribution cost other than depreciation on office building	3362.50		1
Operating profit		1000.00	0.5
Finance cost			
Interest on long term debt		50.00	1
Net profit before tax		950.00	
(-) Taxes		NIL	

Profit after tax transferred to reserves		950.00	0.5
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[12]

ii)

Statement of Financial Position as on 31st March 2014:

	INR million	INR million	Marks
ASSETS			
Non current assets		8277.50	
Office building	712.50		1
Machinery	7565.00		1
Investments in shares of Manchester United Ltd.		222.50	1
Current Assets		3000.00	1
Trade receivables	1900.00		1
Inventories	1050.00		0.5
Cash & bank balance	50.00		1
TOTAL ASSETS		11500.00	
EQUITY AND LIABILITIES			
Share capital	1600.00		1
Reserves and Retained earnings	7900.00		0.5
Total equity		9500.00	
Non current liabilities		500.00	1
Long term debt	500.00		
Current liabilities		1500.00	
Trade payables	1200.00		0.5
Outstanding expenses	300.00		0.5
Total liabilities		2000.00	
TOTAL EQUITY AND LIABILITIES		11500.00	

[10]

Working notes:

- I)** Retained Earnings = Reserves at 31st March 2014 (less) Reserves at 31st March 2013
= 7900 – 6950
= 950 million.

In the absence of taxes and dividends,

Profit after Tax = Net profit before tax = Retained earnings

- II)** Interest cover = 20 times. Let Interest be 'x' million

$$(950 + x) \div x = 20$$

Hence x = Interest paid = INR 50 million

Therefore Operating profit = Net profit before tax + Interest paid
= 950 + 50 = INR 1000 million

III) Gross profit : Operating profit = 4.4:1

$$\text{Gross profit} = 4.4 \times 1000 = \text{INR } 4400 \text{ million}$$

IV) Gross profit margin = 23.15789%

$$\text{Therefore Sales} = 4400 \div 23.15789\% = 19000.0039 \text{ million rounded to INR } 19000 \text{ million}$$

$$\text{Cost of Sales} = \text{Sales Revenue} - \text{Gross Profit} = 19000 - 4400 = \text{INR } 14600 \text{ million}$$

V) Creditors turnover period = 36.5 days and 100% of Purchases are on credit. Also Trade Payables = INR 1200 million

$$\text{i.e. } (\text{Trade Payables} \div \text{Credit Purchases}) \times 365 = 36.5$$

$$\text{Credit Purchases} = \text{Total purchases} = \text{INR } 12000 \text{ million}$$

VI) Let inventory as on 31st March 2014 be INR 'y' million

$$\text{Inventory turnover period} = 17.5 \text{ days}$$

$$\text{i.e. } \{[(y + 350)/2] \div \text{Cost of sales}\} \times 365 = 17.5$$

$$\text{i.e. } \{[(y + 350)/2] \div 14600\} \times 365 = 17.5 \text{ Hence } y = \text{INR } 1050 \text{ million}$$

VII) Cost of Sales = Opening Inventory + Purchases + Depreciation on Machinery + Manufacturing Expenses other than depreciation on Machinery – Closing Inventory

$$\text{i.e. } 14600 = 350 + 12000 + \text{Depreciation on Machinery} + 1965 - 1050$$

$$\text{Hence Depreciation on Machinery} = \text{INR } 1335 \text{ million}$$

VIII) Depreciation on office building = Total depreciation – Depreciation on machinery
= 1372.50 – 1335 = INR 37.50 million

$$\begin{aligned} \text{Total Administrative expenses and Distribution costs} &= \text{Gross Profit} - \text{Operating Profit} \\ &= 4400 - 1000 = \text{INR } 3400 \text{ million} \end{aligned}$$

$$\begin{aligned} \text{Administrative expenses and Distribution costs other than depreciation on office building} &= \\ 3400 - 37.5 &= \text{INR } 3362.50 \text{ million} \end{aligned}$$

IX) Depreciation on machinery = 15% on balance as at 31st March 2013.

$$\text{Balance of machinery as at 31st March 2013} = 1335 \div 15\% = \text{INR } 8900$$

$$\text{Hence Balance of machinery as at 31st March 2014} = 8900 - 1335 = \text{INR } 7565 \text{ million.}$$

Similarly

$$\text{Depreciation on office building} = 5\% \text{ on balance as at 31st March 2013.}$$

$$\text{Balance of office building as at 31st March 2013} = 37.50 \div 5\% = \text{INR } 750.00$$

$$\text{Hence Balance of office building as at 31st March 2014} = 750 - 37.50 = \text{INR } 712.50 \text{ million.}$$

X) Debtors Turnover Period = 50 days. Credit sales = 73% of total sales.

$$\text{Hence Credit sales} = 73\% \text{ of } 19000 = \text{INR } 13870$$

Let Trade Receivables be 'z' million

$$(z \div 13870) \times 365 = 50$$

$$z = \text{Trade Receivables} = \text{INR } 1900 \text{ million}$$

XI) Current Ratio = 2

i.e. $(\text{Inventory} + \text{Trade Receivables} + \text{Cash and bank balance}) \div (\text{Trade payables} + \text{Outstanding expenses}) = 2$
 $(1050 + 1900 + \text{Cash and bank balance}) \div (1200 + 300) = 2$
 Cash and bank balance = INR 50 million

XII) ROCE = 10%

Net profit before tax $\div (\text{Share capital} + \text{reserves}) = 10\%$
 $950 \div (\text{Share capital} + 7900) = 10\%$
 Hence Share capital = INR 1600 million

XIII) Asset utilisation ratio = 1.90

Revenue $\div (\text{Share capital} + \text{Reserves} + \text{Long term debt}) = 1.90$
 $19000 \div (1600 + 7900 + \text{Long term debt}) = 1.9$
 Long term debt = INR 500 million

The balancing figure on the asset side is the value of investment made in Manchester United Ltd.

[22 Marks]

Solution 12:

i) Depreciation is defined as the measure of the wearing out, consumption or other reduction in the useful economic life of a non-current asset, whether arising from:

- effluxion (passage) of time, or
- obsolescence through technological or market changes.

[2]

ii) The two main methods for charging depreciation are as follows –

1. The straight line basis
2. The reducing balance method

[3]

Formula for calculating the rate of depreciation to be charged under each of the above methods-

1. The straight line basis :

$$\frac{(\text{Cost} - \text{Estimated residual value})}{(\text{Estimated useful life})} \times \frac{100}{\text{Cost}}$$

2. The reducing balance method:

$$1 - \sqrt[n]{(\text{Estimated Residual Value}) \div \text{Cost}}$$

Where n – Estimated Useful Life in years

iii) The reducing balance method helps a company to report lower profits in the initial years.

[1]

[6 Marks]

Solution 13:

- i) Maximising the returns to shareholders is the primary objective of a Finance Manager. [1]
- ii) Factors that influence his decisions while trying to achieve the above objective:
- a) the variability of anticipated returns (having regard to the nature of the business)
 - b) the owners' desire for immediate profit rather than future high growth
 - c) the risk appetite of the owners of the business
 - d) the willingness (or otherwise) of the owners to put additional capital into the business
 - e) their willingness to see a reduction in the proportion of the business which they own
- [2]
- iii) Factors that influence the decision on how an asset is financed:
- a) The acceptable degree of gearing
 - b) The effect of taxation.
 - c) The nature of business as also the volatility of its cashflows and profits
- [1]
[4 Marks]

Solution 14:

- i)
Mr Kohli is exposed to the following market risks:
- 1) Rise in the price of shares of Anushka Ltd
 - 2) Appreciation of the rupee against the dollar
- [1]

ii)

Risk of Rise in price of shares of Anushka Ltd.

Mr Kohli could buy a call option on the shares of Anushka Ltd maturing in 3 months' time

Advantage: Mr Kohli could benefit from the fall in the price of shares of Anushka Ltd after 3 months. So he is protected from downside but could benefit from the upside

Drawback: Mr Kohli needs to pay a premium at the outset.

Mr Kohli could also buy a futures contract on the shares of Anushka Ltd maturing in 3 months' time

Advantage:

No premium needs to be paid.

There is no credit risk as exchange is the counterparty to the trade.

Drawback:

Mr Kohli can't benefit from any fall in the price of shares

Appreciation of the rupee against the dollar

Mr Kohli could enter into a forward contract to sell the \$195,950 in a month's time

Advantages:

Since the amount receivable is odd, Mr Kohli would not be able to hedge it exactly using the futures contracts on dollars. Thus a forward contract can be tailor-made to meet one's requirements.

Drawbacks:

Since the forward contracts are traded OTC, Mr Kohli will be exposed to credit risk.
Mr Kohli will not be able to benefit from appreciation of dollar against the rupee

Mr. Kohli could buy suitable number of put options on dollars maturing in a month's time.

Advantages:

Mr Kohli could benefit from the appreciation of dollar in a month's time. So he is protected from downside but could benefit from the upside

Drawbacks:

Mr Kohli needs to pay a premium at the outset.

[9]

[10 Marks]

Solution 15:**1. Swap**

A swap is a contract...
between two parties...
under which they agree to exchange a series of payments...
according to a pre-arranged formula.

2. Warrant

A warrant is an option...
issued by a company...
The holder has a right to repurchase shares...
at a specified price at specified times in future.

[4 Marks]

Solution 16:

Agency costs refer to the problems created in an organisation when there are conflicts in the objectives of different stakeholders (owners and managers)

Agency costs are incurred when:

- managers (as agents) do not attempt to maximize the value of the company
- shareholders (as principals) incur costs monitoring the managers and attempting to influence their actions.

[4 Marks]

Solution 17:

i. In order to calculate beta factor, we need Covariance of Investment and Market Portfolio and Variance of market portfolio, for which we also need to calculate expected returns on market as well as investment portfolio.

Let R_i be Expected return on Investment Portfolio

And R_m be Expected return on Market Portfolio

$$R_i = 1/3 * 6\% + 1/3 * 30\% + 1/3 * 18\% \\ = 18\%$$

$$R_m = 1/3 * 9\% + 1/3 * 12\% + 1/3 * 18\% \\ = 13\%$$

Probability	$R_m - \text{Avg. } R_m$	$R_i - \text{Avg } R_i$	$P(R_m - \text{Avg } R_m)^2$	$P(R_m - \text{Avg } R_m)(R_i - \text{Avg } R_i)$
1/3	-4	-12	5.333	16
1/3	-1	12	0.333	-4
1/3	5	0	8.333	0
			14.00	12

$$\text{Variance of Market Portfolio} = P(R_m - \text{Avg } R_m)^2 = 14$$

$$\text{Covariance of Investment and Market Portfolio} = P(R_m - \text{Avg } R_m)(R_i - \text{Avg } R_i) = 12$$

$$\text{Beta of Investment} = \text{COV } (i,m) / \text{Var}(M) \\ = 12/14 \\ = 0.8571$$

Return required from investment based on its level of systematic risk can be ascertained by CAPM

$$\begin{aligned} \text{Required Return on } R_i &= \text{Interest free rate} + (\text{Market Return} - \text{Interest free rate}) * \text{Beta} \\ &= 6\% + (13\% - 6\%) * 0.8571 \\ &= 6\% + 6\% \\ &= 12\% \end{aligned}$$

The expected return on Investment (18%) is more than the required rate of return (12%) thus, the investment can be accepted.

[8]

ii. In the second case

$$R_i = 2/3 * 6\% + 1/6 * 30\% + 1/6 * 18\% \\ = 12\%$$

$$R_m = 2/3 * 9\% + 1/6 * 12\% + 1/6 * 18\% \\ = 11\%$$

Probability	$R_m - \text{Avg. } R_m$	$R_i - \text{Avg } R_i$	$P(R_m - \text{Avg } R_m)^2$	$P(R_m - \text{Avg } R_m)(R_i - \text{Avg } R_i)$
2/3	-2	-6	2.667	8
1/6	1	18	0.167	3
1/6	7	6	8.167	7
			11.00	18

$$\text{Variance of Market Portfolio} = P(R_m - \text{Avg } R_m)^2 = 11$$

$$\text{Covariance of Investment and Market Portfolio} = P(R_m - \text{Avg } R_m)(R_i - \text{Avg } R_i) = 18$$

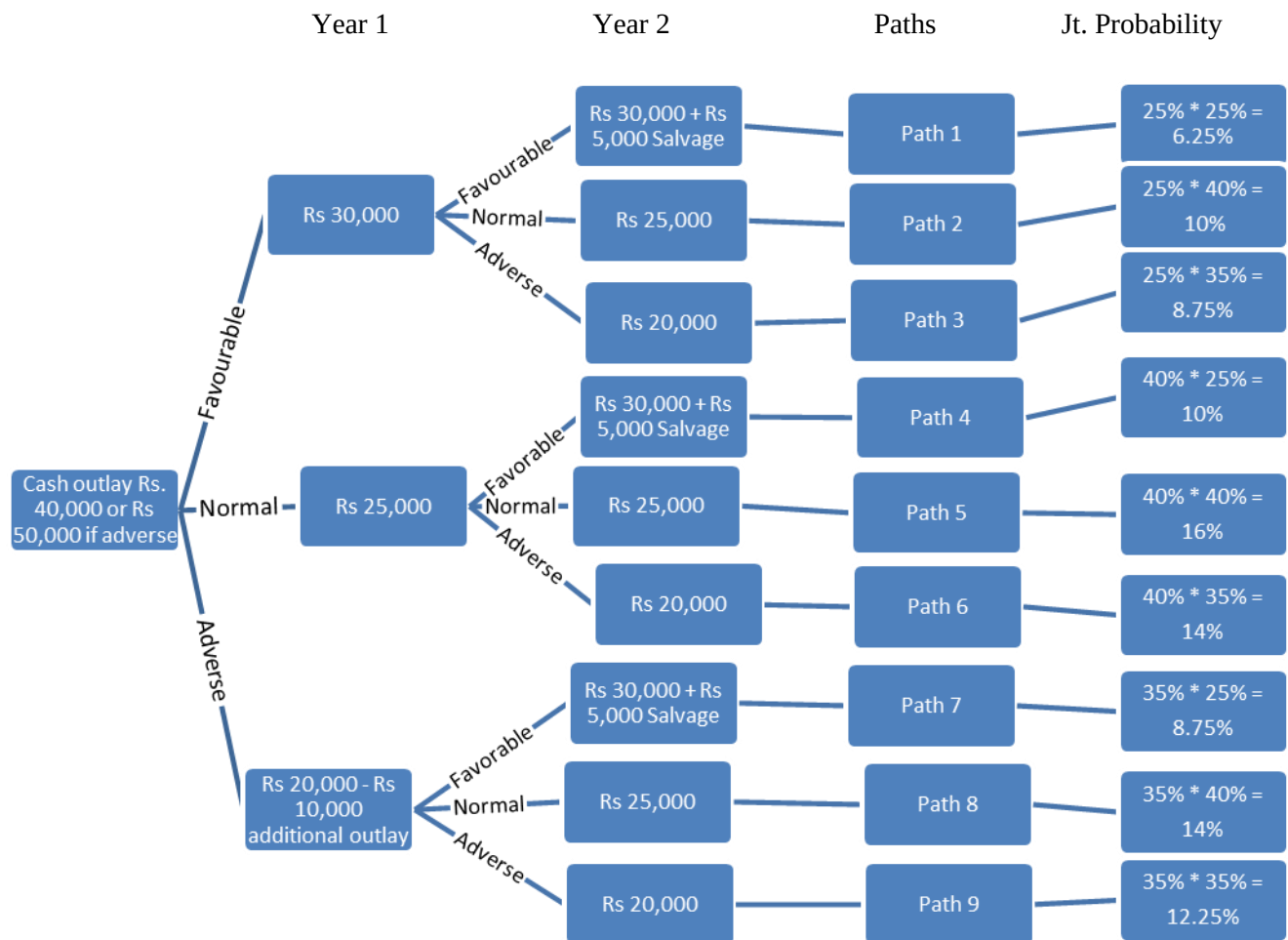
$$\begin{aligned} \text{Beta of Investment} &= \text{COV } (i,m) / \text{Var}(M) \\ &= 18/11 \\ &= 1.6364 \end{aligned}$$

$$\begin{aligned}
 \text{Required Return on Ri} &= \text{Interest free rate} + (\text{Market Return} - \text{Interest free rate}) * \text{Beta} \\
 &= 6\% + (11\% - 6\%) * 1.6364 \\
 &= 6\% + 8.182\% \\
 &= 14.182\%
 \end{aligned}$$

The expected return on Investment (12%) is less than the required rate of return (14.182%) thus, the investment cannot be accepted.

[4]

[12 Marks]

Solution 18:**i) Decision tree**

Statement showing Net Present Value for each path

Paths	Cashflow in year 0	PV factor	Cashflow at end of year 1	PV Factor	Cashflow at end of year 2	PV factor	Net Present Value	Joint Probability	Expected Net Present Value
1	-40000	1	30000	0.9091	35000	0.8264	16198	6.25%	1,012.4
2	-40000	1	30000	0.9091	25000	0.8264	7934	10.00%	793.4
3	-40000	1	30000	0.9091	20000	0.8264	3802	8.75%	332.6
4	-40000	1	25000	0.9091	35000	0.8264	11653	10.00%	1,165.3
5	-40000	1	25000	0.9091	25000	0.8264	3388	16.00%	542.1
6	-40000	1	25000	0.9091	20000	0.8264	-744	14.00%	- 104.1
7	-50000	1	10000	0.9091	35000	0.8264	-11983	8.75%	- 1,048.6
8	-50000	1	10000	0.9091	25000	0.8264	-20248	14.00%	- 2,834.7
9	-50000	1	10000	0.9091	20000	0.8264	-24380	12.25%	- 2,986.6
								NPV	-3,128.1

[12]

- ii) The worst outcome is for Path 9, the Net Present Value for this path is Rs. -24,380 and the probability of this path is 12.25%

[2]

- iii) Paths 1 to 5 have positive NPV while Paths 6 to 9 have negative NPV. The probability of getting positive NPV is 51% (sum of Joint probabilities of path 1 to 5)

[2]

- iv) The project **will not** be accepted since the expected Net Present Value is negative at -3,128.1.

[2]

[18 Marks]
